

SMARTBRANCH 360

Secure Branch Office Network Using Cisco Packet Tracer and Python

PROJECT REPORT

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Course: Networking Essentials

Domain: Networking · Cisco Packet Tracer · Python

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1. Project Overview

Project Name: SmartBranch 360

Domain: Networking, Cisco Packet Tracer, and Python

Main Course: Networking Essentials

Tools

- Cisco Packet Tracer
- Python 3
- YAML
- Terminal / command-line tools

Project Type: Secure branch-office network design, implementation, troubleshooting, and configuration assurance.

The purpose of SmartBranch 360 is to design and implement a secure small-office network that supports:

- Employee wired connectivity
- Guest wireless connectivity
- Internal server access
- Internet connectivity
- DHCP
- DNS reachability
- Inter-VLAN routing
- NAT
- Network-device SSH management
- Guest network isolation
- Configuration validation using Python

The Cisco problem statement specifically requires the network to be designed in Packet Tracer, tested under normal operation, intentionally broken for troubleshooting, and supported by a Python configuration-assurance tool.

2. Problem Statement

A company is opening a new branch office and requires a reliable and secure network infrastructure. The branch needs:

- Wired employee access
- Guest Wi-Fi

- An internal server
- Internet access
- Secure management of network devices
- Separation of different types of network traffic
- A method of validating the configuration automatically

The network must therefore provide both connectivity and security.

The project also requires deliberate fault injection so that troubleshooting can be demonstrated rather than simply showing a working configuration.

3. Project Objectives

Objective 1 — Network Design

Design a small branch-office topology using Cisco Packet Tracer.

Objective 2 — VLAN Segmentation

Create separate logical networks for:

- Employees
- Guests
- Servers
- Management

Objective 3 — IP Addressing

Assign a separate subnet and gateway to each VLAN.

Objective 4 — Network Services

Provide:

- DHCP per VLAN
- DNS reachability
- NAT
- Inter-VLAN routing

Objective 5 — Security

Restrict guest access to internal networks and permit secure SSH management only from the management network.

Objective 6 — Troubleshooting

Create at least five intentional faults and demonstrate their diagnosis and correction.

Objective 7 — Automation

Develop a Python checker that reads the requirements plan and/or Cisco show command output and identifies configuration problems.

These objectives directly correspond to the official workflow and requirements.

4. Network Requirements

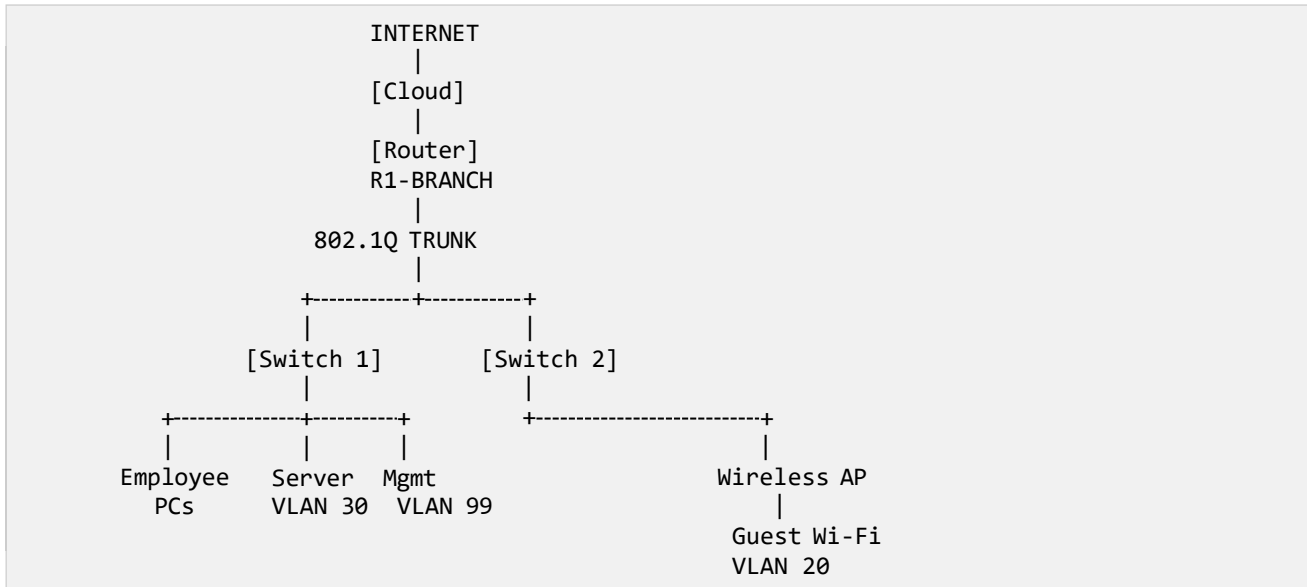
According to the project statement, the topology must contain:

Component	Requirement
Router	1
Switches	2
Wireless Access Point	1
Server	1
Internet / Cloud	1
Endpoints	8 or more

The project also requires four logical VLANs: Employee, Guest, Server, and Management.

5. Proposed Network Architecture

A logical representation of the SmartBranch network is shown below.



The exact physical arrangement should match the final Packet Tracer file.

6. VLAN Design

The project uses four VLANs.

VLAN Table

VLAN ID	VLAN Name	Purpose	Network	Default Gateway
10	EMPLOYEE	Employee devices	10.10.10.0/24	10.10.10.1
20	GUEST	Guest / Wi-Fi devices	10.10.20.0/24	10.10.20.1
30	SERVER	Internal server resources	10.10.30.0/24	10.10.30.1
99	MANAGEMENT	Network administration	10.10.99.0/24	10.10.99.1

The Python requirements checker currently confirms all four VLAN names, subnets, gateways, and gateway/subnet relationships.

7. IP Addressing Plan

VLAN 10 — Employee

- Network: 10.10.10.0/24
- Gateway: 10.10.10.1
- Purpose: Employee computers

Employee computers receive their IP configuration using DHCP.

VLAN 20 — Guest

- Network: 10.10.20.0/24
- Gateway: 10.10.20.1
- Purpose: Guest wireless / client devices

Guest clients receive IP configuration through DHCP. The Guest VLAN is isolated from internal resources.

VLAN 30 — Server

- Network: 10.10.30.0/24
- Gateway: 10.10.30.1
- Purpose: Internal server network

The internal server is placed in the Server VLAN.

VLAN 99 — Management

- Network: 10.10.99.0/24
- Gateway: 10.10.99.1
- Purpose: Network management

The management network is used for secure administration of network devices.

8. Gateway Validation

Each gateway must belong to its corresponding subnet.

VLAN	Gateway	Subnet	Result
10	10.10.10.1	10.10.10.0/24	PASS
20	10.10.20.1	10.10.20.0/24	PASS
30	10.10.30.1	10.10.30.0/24	PASS
99	10.10.99.1	10.10.99.0/24	PASS

The Python checker reports all four gateway/subnet relationships as valid.

9. VLAN Roles

VLAN	Role	Access Policy
10	Employee	Normal business network access
20	Guest	Internet access; internal networks restricted
30	Server	Internal services
99	Management	Network administration and SSH

10. Inter-VLAN Routing

Inter-VLAN routing is required because the four VLANs use separate IP networks. The router provides the Layer-3 gateways:

- VLAN 10 → 10.10.10.1
- VLAN 20 → 10.10.20.1
- VLAN 30 → 10.10.30.1
- VLAN 99 → 10.10.99.1

The router uses subinterfaces for the VLAN networks if implementing router-on-a-stick. Conceptually:

```
R1
|
+-- VLAN 10 -> 10.10.10.1
|
+-- VLAN 20 -> 10.10.20.1
|
+-- VLAN 30 -> 10.10.30.1
|
+-- VLAN 99 -> 10.10.99.1
```

The requirements checker reports:

```
[PASS] inter-VLAN routing requirement is enabled
```

The final Packet Tracer implementation still needs to be demonstrated through actual connectivity tests.

11. Trunking

The switch-to-router and/or switch-to-switch links carrying multiple VLANs use trunking. The trunk needs to carry:

- VLAN 10
- VLAN 20

- VLAN 30
- VLAN 99

This is especially important for Guest VLAN 20 because the official Cisco example identifies a missing VLAN on a trunk as a troubleshooting scenario.

12. DHCP Design

Each VLAN has its own DHCP pool.

DHCP Pool Table

DHCP Pool	VLAN	Network	Gateway
EMPLOYEE	10	10.10.10.0/24	10.10.10.1
GUEST	20	10.10.20.0/24	10.10.20.1
SERVER	30	10.10.30.0/24	10.10.30.1
MANAGEMENT	99	10.10.99.0/24	10.10.99.1

The real `show_dhcp.txt` output contains:

```
Pool EMPLOYEE
Pool GUEST
Pool SERVER
Pool MANAGEMENT
```

The modified Python checker correctly recognizes Cisco's "Pool NAME :" output format. The final checker result is:

```
PASS : 47
WARN : 0
FAIL : 0

Overall result: PASS
```

13. DNS

DNS reachability is one of the required services. The purpose of DNS is to allow clients to resolve host/domain names instead of requiring users to remember IP addresses. The project specifically requires DNS reachability as part of the services.

Required Test

From an appropriate client:

```
DNS resolution -> SUCCESS
```

This should be demonstrated in Packet Tracer.

14. NAT

NAT allows internal private addresses to communicate with the Internet. The internal networks are:

- 10.10.10.0/24
- 10.10.20.0/24
- 10.10.30.0/24
- 10.10.99.0/24

NAT translates appropriate internal traffic toward the Internet connection. The project explicitly requires NAT for Internet access.

15. Security Architecture

The network uses segmentation and ACL-based restrictions.

Security Requirements

Source	Destination	Required Result
Guest VLAN 20	Server VLAN 30	DENY
Guest VLAN 20	Management VLAN 99	DENY
Guest VLAN 20	Employee VLAN 10	DENY
Guest VLAN 20	Internet	ALLOW
Management VLAN 99	Network devices via SSH	ALLOW

The official project specifically requires guest traffic to be blocked from the server VLAN and SSH management to be restricted to the management VLAN.

16. ACL Design

The required ACL logic includes:

```
deny Guest -> Server
deny Guest -> Management
deny Guest -> Employee
permit Guest -> any
```

The Python checker currently confirms:

```
[PASS] Guest -> Server      = DENY
[PASS] Guest -> Management = DENY
[PASS] Guest -> Employee   = DENY
```

and also validates the corresponding ACL rules from show_acl.txt.

17. SSH Security

Network devices should support secure remote administration. The requirement is:

- Management host → SSH → ALLOW
- Other networks → SSH → RESTRICTED

The Python checker validates SSH version 2 and checks for the following, where present in the show output:

- login local
- transport input ssh

The requirements checker reports:

```
[PASS] Management host -> SSH = ALLOW
```

18. End-to-End Traffic Requirements

The final Packet Tracer network should demonstrate the following.

Employee

```
Employee PC
|
+---- DHCP      -> SUCCESS
+---- Server    -> SUCCESS
+---- Internet  -> SUCCESS
```

The assignment explicitly states that employee PCs should obtain DHCP, reach the server, and reach the Internet.

Guest

```
Guest PC
|
+---- DHCP      -> SUCCESS
+---- Internet  -> SUCCESS
+---- Server    -> BLOCKED
+---- Management -> BLOCKED
```

The official pass condition is that Guest reaches the Internet but not the server or management network.

Management

```
Management PC
|
+---- SSH to router  -> SUCCESS
+---- SSH to switches -> SUCCESS
```

Only the management host/network should have administrative SSH access.

19. Python Configuration Assurance Tool

File: SmartBranch_Python_Checker.py

The Python tool is one of the core deliverables. The project requires a Python tool that reads a plan or Cisco show outputs and reports likely configuration problems. The tool validates:

- VLAN names
- VLAN subnets
- VLAN gateways
- Gateway/subnet relationships
- DHCP requirement
- DNS requirement
- NAT requirement
- Inter-VLAN routing requirement
- ACL rules
- DHCP pools
- Interface gateway addresses
- SSH configuration
- Trunk configuration
- VLAN output

20. Python Tool Input

The project uses:

```
SmartBranch_Requirements.yaml
SmartBranch_Real_Show_Outputs/
```

The real show-output directory contains:

- show_acl.txt
- show_dhcp.txt
- show_interfaces.txt
- show_ssh.txt
- show_trunk.txt
- show_vlan_brief.txt

21. Python Tool Output

The final validation produces:

```
=====
SMARTBRANCH 360 VALIDATION SUMMARY
=====

PASS : 47
WARN : 0
FAIL : 0

Overall result: PASS
```

No required configuration checks failed. This satisfies the project's automation requirement that the Python tool produce useful and readable validation output.

22. Requirements Validation Summary

The current requirements-plan validation is:

Requirement	Result
VLAN 10 name	PASS
VLAN 10 subnet	PASS
VLAN 10 gateway	PASS
VLAN 10 gateway/subnet relationship	PASS
VLAN 20 name	PASS

Requirement	Result
VLAN 20 subnet	PASS
VLAN 20 gateway	PASS
VLAN 20 gateway/subnet relationship	PASS
VLAN 30 name	PASS
VLAN 30 subnet	PASS
VLAN 30 gateway	PASS
VLAN 30 gateway/subnet relationship	PASS
VLAN 99 name	PASS
VLAN 99 subnet	PASS
VLAN 99 gateway	PASS
VLAN 99 gateway/subnet relationship	PASS
DHCP	PASS
DNS	PASS
NAT	PASS
Inter-VLAN routing	PASS
Guest → Server	PASS
Guest → Management	PASS
Guest → Employee	PASS
Management → SSH	PASS

24 / 24 requirements passed.

23. Show-Output Validation

The current real-output checker result is:

Result	Count
PASS	47
WARN	0
FAIL	0

Overall result: PASS

This is an important result to show in the project demonstration.

24. Fault-Tolerant Troubleshooting

A major part of SmartBranch 360 is intentionally breaking the network. The official requirement states that the network must be broken at least five times, and gives examples including a wrong gateway, a missing VLAN on a trunk, bad DHCP, an ACL blocking DNS, and NAT problems.

25. Fault Scenario 1 — Wrong Default Gateway

Fault

Change an employee PC's gateway from 10.10.10.1 to an incorrect address, for example 10.10.10.254.

Symptom

The employee PC may receive an IP address but cannot communicate correctly outside its local network.

Root Cause

Incorrect default gateway.

Evidence

Check ipconfig and compare the gateway against the VLAN design.

Fix

Set Default Gateway = 10.10.10.1

Verification

```
ping 10.10.10.1
ping 10.10.30.1
```

Also test server and Internet connectivity.

26. Fault Scenario 2 — VLAN 20 Missing from Trunk

Fault

Remove VLAN 20 from the trunk's allowed VLAN list.

Symptom

Guest PC loses connectivity.

Root Cause

VLAN 20 cannot traverse the trunk.

Evidence

```
show interfaces trunk
```

The VLAN 20 entry is missing. The Cisco assignment itself gives this type of failure as its example Python finding.

Fix

Add VLAN 20 to the trunk.

Verification

Guest PC should:

- Obtain DHCP
- Reach Internet
- Remain blocked from internal networks

27. Fault Scenario 3 — DHCP Failure

Fault

Break or remove a DHCP pool.

Symptom

A client cannot obtain a valid IP address.

Root Cause

Missing or incorrect DHCP configuration.

Evidence

```
show ip dhcp pool
```

Or inspect the client IP configuration.

Fix

Restore the appropriate DHCP pool, for example:

- EMPLOYEE
- GUEST
- SERVER
- MANAGEMENT

Verification

Set the PC to DHCP and confirm it receives an address in the correct subnet.

28. Fault Scenario 4 — ACL Blocks DNS

Fault

Add an ACL rule that unintentionally blocks DNS traffic.

Symptom

The client may have network connectivity but cannot resolve names.

Root Cause

DNS traffic is being denied by an ACL.

Evidence

```
show access-lists
```

And inspect DNS connectivity.

Fix

Correct the ACL so legitimate DNS traffic is permitted.

Verification

Test DNS resolution again.

29. Fault Scenario 5 — NAT Failure

Fault

Break the NAT configuration.

Symptom

Internal clients can reach the router/internal network but cannot access the Internet.

Root Cause

Incorrect or missing NAT configuration.

Evidence

```
show ip nat translations
show ip nat statistics
```

Fix

Restore the correct NAT configuration.

Verification

Test Internet access from an employee or guest client.

30. Fault Card Table

For the final submission, the five fault cards can be summarized as follows:

Fault	Symptom	Root Cause	Fix
Wrong Gateway	No external connectivity	Incorrect gateway	Restore correct VLAN gateway
Missing VLAN on Trunk	Guest connectivity fails	VLAN 20 not allowed on trunk	Add VLAN 20
DHCP Failure	Client gets no valid IP	DHCP pool/configuration problem	Restore DHCP pool
ACL Blocks DNS	Name resolution fails	DNS traffic denied	Correct ACL
NAT Failure	No Internet access	NAT misconfiguration	Correct NAT

31. Testing Plan

The project should be tested in five major categories.

Test 1 — VLAN / DHCP

Test	Expected
Employee PC DHCP	PASS
Guest PC DHCP	PASS
Server VLAN addressing	PASS
Management addressing	PASS

Test 2 — Employee Connectivity

Test	Expected
Employee → Gateway	PASS
Employee → Server	PASS
Employee → Internet	PASS

Test 3 — Guest Isolation

Test	Expected
Guest → Gateway	PASS
Guest → Internet	PASS
Guest → Server	BLOCK
Guest → Management	BLOCK
Guest → Employee	BLOCK

Test 4 — Secure Management

Test	Expected
Management PC → Router SSH	PASS
Management PC → Switch SSH	PASS
Guest → Device SSH	BLOCK
Employee → Device SSH	BLOCK

The official checking criteria specifically require that only the management host be able to SSH to network devices.

32. Python Validation Test

Run:

```
cd ~/Downloads/SmartBranch_Python_Checker
python3 SmartBranch_Python_Checker.py \
  --plan SmartBranch_Requirements.yaml \
  --show-dir SmartBranch_Real_Show_Outputs
```

Expected final output:

```
PASS : 47
WARN : 0
FAIL : 0
```

Overall result: PASS

No required configuration checks failed.

33. Project Files

The final Python project should contain:

```
SmartBranch_Python_Checker/
|
+-- SmartBranch_Python_Checker.py
+-- SmartBranch_Requirements.yaml
+-- README.md
|
+-- SmartBranch_Real_Show_Outputs/
|   +-- show_acl.txt
|   +-- show_dhcp.txt
|   +-- show_interfaces.txt
|   +-- show_ssh.txt
|   +-- show_trunk.txt
|   +-- show_vlan_brief.txt
|
+-- sample_show_outputs/
|   +-- show_acl.txt
|   +-- show_dhcp.txt
|   +-- show_interfaces.txt
|   +-- show_ssh.txt
|   +-- show_trunk.txt
|   +-- show_vlan_brief.txt
```

The clean ZIP has already been checked with:

```
unzip -l SmartBranch_Python_Checker_FINAL_CLEAN.zip
```

and it contains the expected project files.

34. Final Project Conclusion

SmartBranch 360 provides a secure and segmented branch-office network using Cisco Packet Tracer and Python.

The design separates employees, guests, servers, and management traffic using VLANs and dedicated IP subnets. DHCP provides automatic client addressing, while DNS, NAT, and inter-VLAN routing provide required network services.

Security is implemented by restricting Guest access to internal networks and limiting network-device SSH management to the management network.

The Python configuration-assurance tool automates validation of the requirements and Cisco show-command outputs. The final checker currently reports:

- 47 PASS
- 0 WARN
- 0 FAIL
- Overall result: PASS

The troubleshooting component demonstrates practical network diagnosis by intentionally introducing faults such as incorrect gateways, missing trunk VLANs, DHCP problems, ACL issues, and NAT failures.

The project therefore combines network design, Cisco configuration, network security, troubleshooting, automation, and technical documentation in a single SmartBranch 360 solution.

